Effect Size: Eta-Squared

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Introduction

Eta-squared (η^2) is the standard effect-size measure for analysis of variance: the proportion of total variance in the outcome attributable to a factor. It complements the omnibus F-test, which reports significance, by quantifying magnitude — how much of the variability in the response is explained by the categorical factor of interest. Several variants of eta-squared are widely used in different ANOVA designs: classical η^2 in one-way designs, partial η_p^2 in factorial between-subjects designs, generalised η_G^2 in mixed designs that combine between- and within-subjects factors, and omega-squared ω^2 as an unbiased alternative preferred in small samples. Reporting the appropriate variant alongside the F-test result is now standard in psychology, neuroscience, and clinical-trial publications.

Prerequisites

A working understanding of one-way and factorial ANOVA, the partitioning of sums of squares into factor and error components, and the distinction between between-subjects and within-subjects factors.

Theory

Classical eta-squared is

$$\eta^2 = \frac{\mathrm{SS}_{\mathrm{factor}}}{\mathrm{SS}_{\mathrm{total}}}.$$

Partial eta-squared, for factorial designs, removes other factors' variance from the denominator:

$$\eta_p^2 = \frac{\mathrm{SS}_{\mathrm{factor}}}{\mathrm{SS}_{\mathrm{factor}} + \mathrm{SS}_{\mathrm{error}}}.$$

Generalised eta-squared (Bakeman, 2005), for mixed designs combining within- and between-subjects factors, uses a denominator that depends on the design and is comparable across between- and within-subjects effects.

Omega-squared ω^2 corrects for the upward bias of η^2 in small samples:

$$\omega^2 = \frac{\mathrm{SS}_{\mathrm{factor}} - \mathrm{df}_{\mathrm{factor}} \cdot \mathrm{MSE}}{\mathrm{SS}_{\mathrm{total}} + \mathrm{MSE}}.$$

Cohen's conventional thresholds for η^2 are 0.01 (small), 0.06 (medium), and 0.14 (large).

Assumptions

The same as the underlying ANOVA: independent observations (within a stratum), Normal residuals, homogeneous variances. The eta-squared family does not impose additional assumptions beyond those of the ANOVA from which the sums of squares are computed.

R Implementation

```
library(effectsize)
set.seed(2026)

df <- data.frame(
  A = factor(rep(c("A1", "A2"), each = 30)),
  B = factor(rep(c("B1", "B2", "B3"), 20)),
  y = rnorm(60)
)
df$y <- df$y + 0.5 * (df$A == "A2") + 0.3 * (df$B == "B2") + 0.1 * (df$B == "B3")

fit <- aov(y ~ A * B, data = df)

eta_squared(fit)
eta_squared(fit, partial = TRUE)
eta_squared(fit, generalized = TRUE)
omega_squared(fit)
```

Output & Results

`effectsize::eta_squared()` and `omega_squared()` return point estimates and confidence intervals for each ANOVA term. The output reports the appropriate variant for the requested design, with η_p^2 interpretable as the variance explained by each factor relative to itself plus error, holding other factors constant.

Interpretation

A reporting sentence: "The factorial ANOVA showed a small main effect of factor A on the outcome (partial $\eta_p^2 = 0.06$, 95 % CI 0.01 to 0.21, $F_{1,54} = 3.4$, $p = 0.07$); other terms were small and non-significant. Reporting partial η_p^2 rather than classical η^2 is appropriate for this between-subjects factorial design because each effect's variance share is computed relative to its own error stratum. Omega-squared values were 0.04 and 0.00 respectively, slightly more conservative as expected." Always specify which η^2 variant.

Practical Tips

- Report partial η_p^2 for between-subjects factorial designs; classical η^2 in factorial contexts conflates a factor's variance share with all other factors and is rarely the right summary.
- Use generalised η_G^2 (Bakeman, 2005) for mixed designs that combine within-subjects and between-subjects factors; it gives effect sizes that are directly comparable across the two types of factor, where partial η_p^2 is not.
- Prefer omega-squared ω^2 in small samples (typically $n < 30$ per cell); η^2 is upward-biased in small samples and ω^2 corrects this. Report both when feasible.
- Always report a confidence interval on the effect size, not just the point estimate; CIs on η^2 are non-central-F-based and are reported by `effectsize::eta_squared()` directly.
- Cohen's thresholds (0.01 / 0.06 / 0.14) come from behavioural research and may not be appropriate benchmarks in biomedical or physical-science contexts where larger effects are routinely expected; interpret thresholds in light of substantive expectations rather than treating them as universal.
- For the related $f^2 = \eta^2/(1 - \eta^2)$ effect size used in regression power analysis, the conversion is direct and useful when designing follow-up studies based on existing ANOVA results.

R Packages Used

`effectsize::eta_squared()`, `effectsize::omega_squared()`, and `effectsize::epsilon_squared()` for the canonical effect-size family with confidence intervals; `lsr::etaSquared()` for an alternative interface;

`MOTE::eta.full.SS()` for textbook-companion calculations; `BayesFactor` for Bayesian effect-size estimation; `Superpower` for ANOVA-design power simulation that also produces eta-squared estimates.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation